Hybrid Monte Carlo Methods for Particle Transport



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Slides & paper: johanneskrotz.org/downloads

Problem

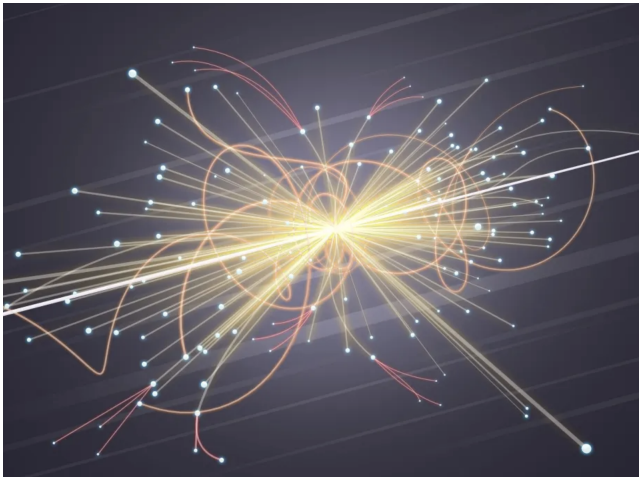
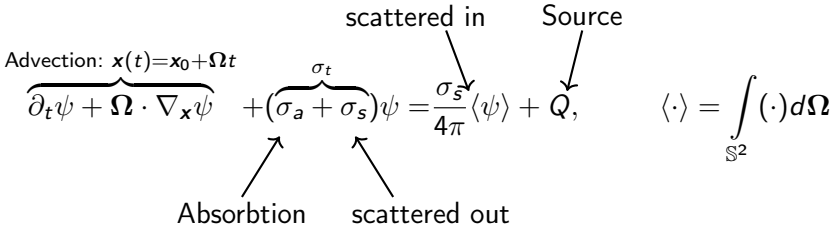
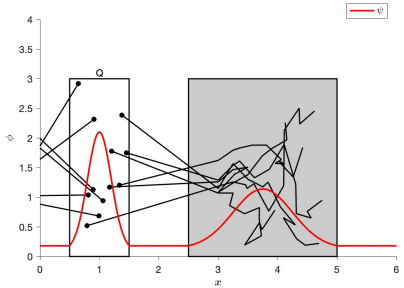


Image source: https://bigthink.com/starts-with-a-bang/attoseconds-particle-physics/



- $\psi = \psi(\boldsymbol{x}, \boldsymbol{\Omega}, t)$ particle density at $\boldsymbol{x} \in \mathbb{R}^2$ flying in direction $\boldsymbol{\Omega} \in \mathbb{S}^2$, $\langle\psi\rangle(\boldsymbol{x}, t)$ angular average
- common approaches:
  - Discontinuous Galerkin (DG)
  - Monte Carlo (MC)
- Hybrid

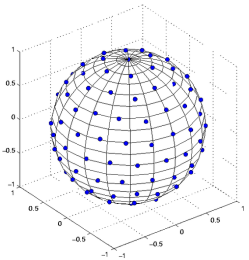


- Very accurate, if particle number is high: $Error \sim \frac{1}{\sqrt{N}}$, while $\mathcal{O}(N)$
- embarrassingly parallel

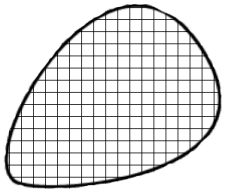
- exploration can be slow
- expensive in high scattering
- QMC not straightforward

- Discretize in
  - time: $t_0, t_1, t_2, ..., t_n$(implicit Euler)
  - direction: $\mathbf{\Omega}_1, \mathbf{\Omega}_2, ...$(discrete Ordinates)
  - space: $\psi(\boldsymbol{x}, \mathbf{\Omega}_q, t) = \sum_k \alpha^{C,q}(t)_k \varphi^k(\boldsymbol{x})$ for $x \in C$

    (Weak formulation, Legendre basis on Cartesian cells)



$\mathbf{\Omega}_q$



$C_{i,j}$

- Results in linear system for $\alpha(t)$s

$$\mathcal{L}\boldsymbol{\alpha}^{q,C} = \mathcal{A}\boldsymbol{\alpha}^{q,C^{\pm}} + \sigma_s \mathcal{B} \sum_{q'} w_{q'} \boldsymbol{\alpha}^{q',C}$$

- If $\sigma_s \neq 0$ fully coupled $\rightarrow$ Curse of dimensionality
- Iterative solvers work well for intermediate scattering
- ray effects

Hybrid motivation

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Linear Boltzmann / transport equation

$$\underbrace{\partial_t \psi + \boldsymbol{\Omega} \cdot \nabla_{\boldsymbol{x}} \psi}_{L\psi} + \sigma_a \psi = \frac{\sigma_s}{4\pi} \langle \psi \rangle + Q,$$

- $\psi(\boldsymbol{x}, \boldsymbol{\Omega}, t)$ is the angular flux.
- Scalar flux: $\phi(\boldsymbol{x}, t) = \langle \psi \rangle = \int_{S^2} \psi(\boldsymbol{x}, \boldsymbol{\Omega}, t)\, d\boldsymbol{\Omega}$.

Numerical tension

- Monte Carlo resolves directional transport naturally, but has sampling noise.
- Deterministic S_N–DG is structured and efficient for smoother scattered fields, but angular coupling is expensive.
- Hybridization lets MC handle the early part and S_N–DG the smoother tail.

Hybrid split

1st-collision hybrid

$$\psi = \underbrace{\psi_0}_{\text{MC}} + \underbrace{\psi_{>0}}_{S_N\text{–DG}}$$

$$L\psi_0 + \sigma_t\psi_0 = Q,$$
$$L\psi_{>0} + \sigma_t\psi_{>0} = \frac{\sigma_s}{4\pi}\langle\psi_0 + \psi_{>0}\rangle.$$

N_s-collision hybrid

$$\psi = \underbrace{\psi_{<N_s}}_{\text{MC below cap}} + \underbrace{\psi_{N_s}}_{\text{exactly } N_s} + \underbrace{\psi_{>N_s}}_{S_N\text{–DG tail}},$$

$$L\psi_{<N_s} + \sigma_t\psi_{<N_s} = Q + \frac{\sigma_s}{4\pi}\langle\psi_{<N_s}\rangle - P_{(N_s-1\to N_s)},$$
$$L\psi_{N_s} + \sigma_t\psi_{N_s} = P_{(N_s-1\to N_s)},$$
$$L\psi_{>N_s} + \sigma_t\psi_{>N_s} = \frac{\sigma_s}{4\pi}\langle\psi_{>N_s}\rangle + \frac{\sigma_s}{4\pi}\langle\psi_{N_s}\rangle.$$

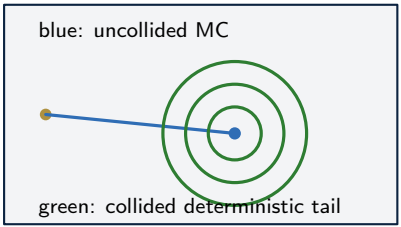
Relabeling: $L\psi_R + \sigma_t\psi_R = Q + \frac{\sigma_s}{4\pi}\langle\psi_{<N_s} + \psi_{N_s} + \psi_{>N_s}\rangle, \qquad \psi(t_{n+1}) \leftarrow \psi_R(t_{n+1})$

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1st-collision hybrid

$$L\psi_0 + \sigma_t\psi_0 = Q$$

$$L\psi_{>0} + \sigma_t\psi_{>0} = \frac{\sigma_s}{4\pi}\langle\psi_0 + \psi_{>0}\rangle$$

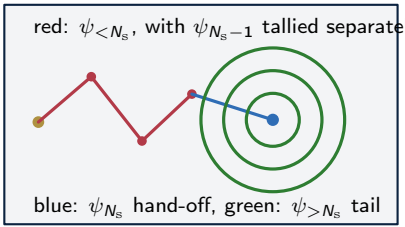


N_s-collision hybrid

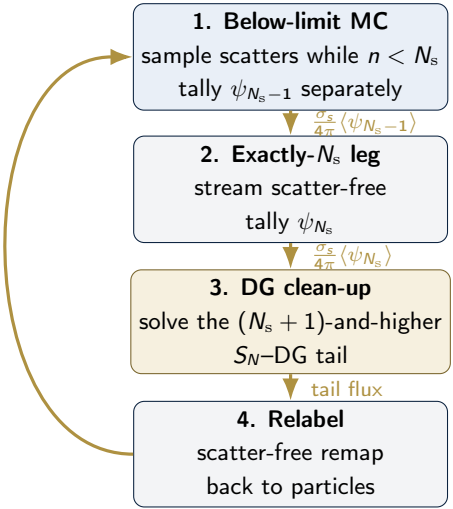
$$L\psi_{<N_s} + \sigma_t\psi_{<N_s} = Q + \frac{\sigma_s}{4\pi}\langle\psi_{<N_s}\rangle - P_{(N_s-1\to N_s)},$$

$$L\psi_{N_s} + \sigma_t\psi_{N_s} = P_{(N_s-1\to N_s)},$$

$$L\psi_{>N_s} + \sigma_t\psi_{>N_s} = \frac{\sigma_s}{4\pi}\langle\psi_{>N_s}\rangle + \frac{\sigma_s}{4\pi}\langle\psi_{N_s}\rangle$$



Hybrid split



Relabeling

Relabeling equation

$$L\psi_R + \sigma_t \psi_R = Q + \frac{\sigma_s}{4\pi} \langle \psi_{\lceil N_s \rceil} + \psi_{N_s} + \psi_{\lfloor N_s+1 \rfloor} \rangle$$

$$\psi(t_{n+1}) \leftarrow \psi_R(t_{n+1})$$

Why relabel?

- Without relabeling, mass keeps drifting into the deterministic tail.
- Then long-time accuracy is increasingly set by the cleanup solve.
- Relabeling resets the next step to an MC representation.

How it is done in practice

1. Assume the full RHS is known from the MC tallies and the S_N–DG solve.
2. Run one *scatter-free* MC transport with attenuation σ_t.
3. Reuse the particles from step 1 by resetting them to their birth data.
4. Copy particles coming from initial/boundary data.

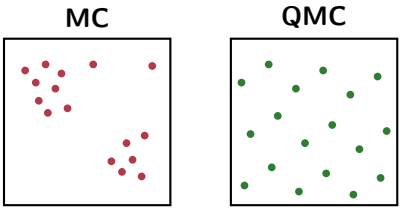
Practical point: no explicit resampling from the DG angular distribution is needed.

General challenge

For fully scattering MC in heterogeneous media, the number of collision events is path dependent. That means the effective stochastic dimension is not fixed in advance, which makes classical QMC constructions awkward.

Why the hybrid helps

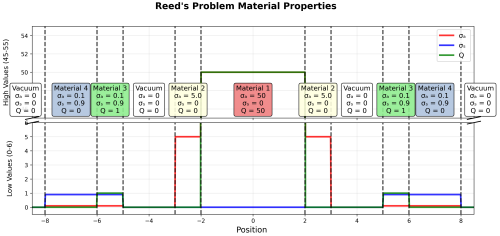
- The relabeling step is scatter-free.
- The implementation can keep the existing inverse-CDF maps and simply replace pseudorandom draws by Halton points.
- In the reported experiments, that change improved convergence in the hybrid setting with little extra cost.



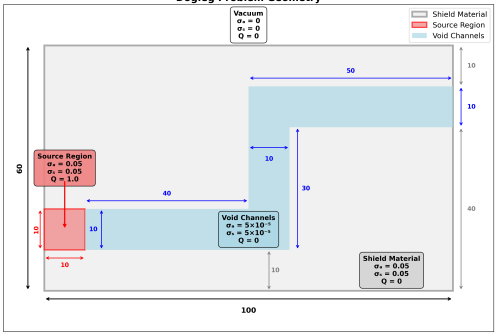
more even coverage
of the sampling space

Benchmarks

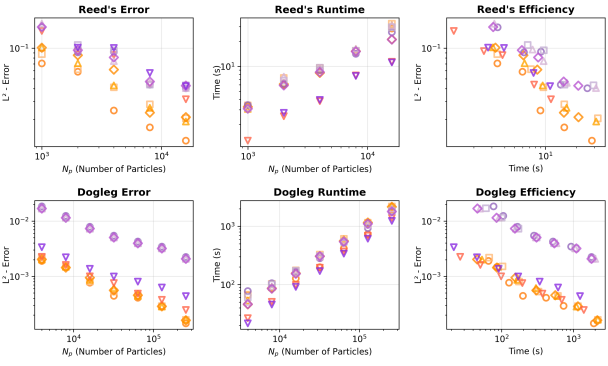
Reed's problem

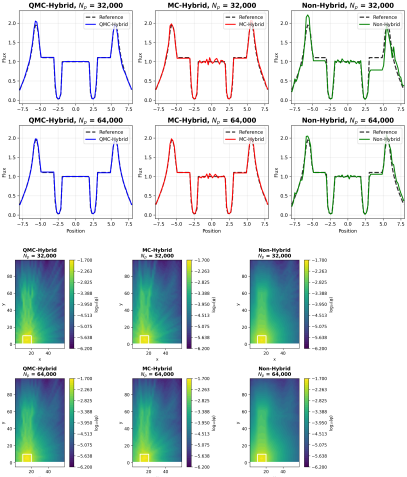


Dogleg problem



Results


Reed's Error
Reed's Runtime
Reed's Efficiency
Dogleg Error
Dogleg Runtime
Dogleg Efficiency
L² - Error
Time (s)
N_p (Number of Particles)
Time (s)
QMC - N_s = 0
MC - N_s = 0
QMC - N_s = 5
MC - N_s = 5
QMC - N_s = 10
MC - N_s = 10
QMC - N_s = 20
MC - N_s = 20
QMC - N_s = ∞
MC - N_s = ∞


QMC-Hybrid, N_p = 32,000
MC-Hybrid, N_p = 32,000
Non-Hybrid, N_p = 32,000
QMC-Hybrid, N_p = 64,000
MC-Hybrid, N_p = 64,000
Non-Hybrid, N_p = 64,000
Reference
QMC-Hybrid
MC-Hybrid
Non-Hybrid
Flux
Position

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Convergence-rate and cleanup-iteration summary

	Reed			Dogleg		
N_s	SN iters	α_{QMC}	α_{MC}	SN iters	α_{QMC}	α_{MC}
0	31	-0.69	-0.50	7	-0.61	-0.49
5	26	-0.64	-0.51	4	-0.61	-0.49
10	21.4	-0.63	-0.53	1	-0.61	-0.49
20	12.2	-0.64	-0.49	1	-0.61	-0.49
∞	0	-0.56	-0.46	0	-0.53	-0.47

Takeaway

- Hybrid + QMC converges faster than hybrid + MC.
- The gain is strongest in the hybrid setting.

Role of N_s

- Larger N_s reduces the SN cleanup burden.
- Accuracy changes mildly; the work split changes.

Takeaways

1. The 1st-collision hybrid becomes an arbitrary scatter-limit hybrid.
2. MC handles the low-scatter, directional part; S_N–DG handles the high-scatter tail.
3. Relabeling is essential: it prevents the deterministic cleanup from taking over the long-time solution quality.
4. Replacing pseudorandom draws by Halton points gives a clear accuracy benefit in the hybrid setting at little extra cost.

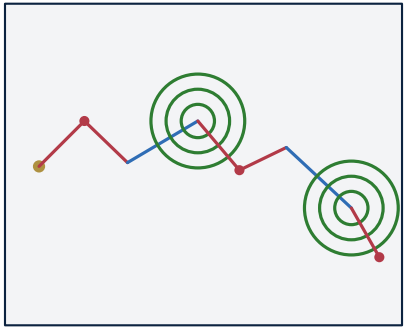
General split

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Labeled MC-DG-Hybrid

$$\psi = \psi_M + \psi_H + \psi_D$$

$$L\psi_M + \sigma_t \psi_M = \frac{\sigma_s}{4\pi}\langle \psi_M \rangle + Q - P_{M\to H}[\psi_M] + P_{H\to M}[\psi_H],$$
$$L\psi_H + \sigma_t \psi_H = P_{M\to H}[\psi_M] - P_{H\to M}[\psi_H],$$
$$L\psi_D + \sigma_t \psi_D = \frac{\sigma_s}{4\pi}\langle \psi_D \rangle + \frac{\sigma_s}{4\pi}\langle \psi_H \rangle.$$



Outlook

- Can N_s be chosen adaptively in space, time, or by local optical thickness / variance indicators?
- How much larger is the payoff in a parallel implementation, where the MC leg is far easier to scale than the coupled cleanup solve?
- Could the cleanup stage be cheaper than S_N–DG in strongly scattering regions?

Questions?